

## Tutorial 1

Instructor: Prof. S Krishna

Date: 5th August 2026

## 1 Do you get déjà vu?

**Q1.** Construct DFA accepting the following languages over the  $\Sigma = \{0, 1\}$

1. Set of all strings where the number of 0's is a multiple of 3
2. Set of all strings ending in 00
3. Set of all strings that begin with a 1 that, when interpreted as a binary integer, are multiples of 5.  
Eg: 101, 1111 are in the language; 0, 100 are not

**Q2.** Draw the NFA that accepts the following languages

1. Set of strings such that final digit has appeared before over the alphabet  $\Sigma = \{0, 1 \dots 9\}$
2. Set of all strings containing exactly one occurrence of the substring 101 over the alphabet  $\Sigma = \{0, 1\}$
3. Set of all strings in which some block of three consecutive symbols contains exactly two 1's over the alphabet  $\Sigma = \{0, 1\}$

**Q3.** A Galton board consists of  $n$  rows of pegs. A ball starts at the top and, at each row, independently moves either left or right before reaching the next row. After passing through all the  $n$  rows, the ball lands in one of the  $n + 1$  bins.

Construct a deterministic finite automaton that accepts precisely those strings that cause the ball to land in the  $k$ -th bin, where  $0 \leq k \leq n$  for  $n = 3, k = 2$ . Your automaton should reject all strings that are not of length  $n$ .

In your construction, specify the alphabet, transition function, start state, accepting states and explain what each state represents. Also think about how the automaton changes as the  $n, k$  change.

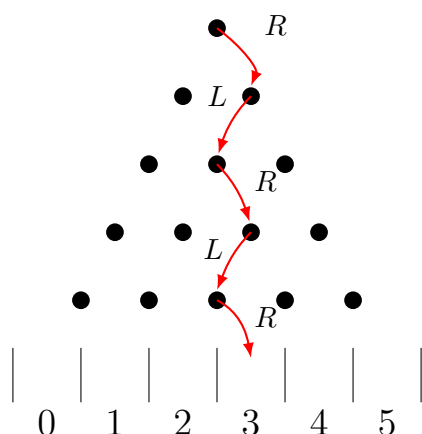


Figure 1: Example of a Galton board instance for  $n = 5, k = 3$

**Q4.** Construct an NFA with  $1n+1$  states that accepts

$$L(n) = \{w \in \{0,1\}^* \mid n\text{-th symbol from the right is } 1\}$$

Also, justify why any equivalent DFA requires at least  $2^n$  states.

**Q5.** Consider a nondeterministic finite automaton  $M = (Q, \Sigma, \delta, q_0, F)$ . We have learnt two notions of acceptance:

- Angelic (existential) acceptance:**  $w \in L_\exists(M)$  iff *some* run of  $M$  on  $w$  ends in  $F$ .
- Devilish (universal) acceptance:**  $w \in L_\forall(M)$  iff *every* run of  $M$  on  $w$  ends in  $F$ .

1. Give an explicit NFA  $M$  such that

$$L_\exists(M) \neq L_\forall(M).$$

Compute both languages and explain in one or two sentences why they differ.

2. Suppose  $M$  is an NFA but it happens that  $L_\exists(M) = L_\forall(M)$  anyway. Does this imply that  $M$  must in fact be a DFA? If yes, prove it. If not, construct a counterexample: an NFA with  $|\delta(q, a)| > 1$  for some  $q \in Q, a \in \Sigma$ , for which  $L_\exists(M) = L_\forall(M)$  still holds, and state what property the branching states must satisfy for this to happen.

**Q6. To Be (Regular), or Not to Be?**

1. Show that regular languages are closed under set difference: if  $L_1$  and  $L_2$  are regular, so is  $L_1 \setminus L_2$ , where  $L_1 \setminus L_2 = \{w \in L_1 : w \notin L_2\}$  denotes the set of strings in  $L_1$  but not in  $L_2$ .
2. Are regular languages closed under infinite union? That is, given regular languages  $L_1, L_2, L_3, \dots$ , is

$$\bigcup_{i=1}^{\infty} L_i$$

always regular?

3. What about closure of regular languages under infinite intersection? That is, is

$$\bigcap_{i=1}^{\infty} L_i$$

always regular? Can you obtain the answer directly from your result in part 2.

**Q7.** Let  $\Sigma = \{a_1, a_2, \dots, a_n\}$  and consider the language

$$L = \{a_1\}^? \{a_2\}^? \dots \{a_n\}^?$$

i.e.,  $L$  consists of all strings formed by taking  $a_1, a_2, \dots, a_n$  in that relative order, where each one is independently optional (may be included or skipped). So, for example, when  $n = 3$

$$L = \{\epsilon, a_1, a_2, a_3, a_1a_2, a_1a_3, a_2a_3, a_1a_2a_3\}$$

1. Draw an  $\epsilon$ -NFA accepting  $L$  using as few states as possible. (*Hint: use  $\epsilon$ -transitions*)
2. Now construct an NFA with **no**  $\epsilon$ -transitions accepting the same language  $L$ .
3. State the number of states and the number of transitions used in each of your two machines from parts (a) and (b), as a function of  $n$ . Does removing  $\epsilon$ -transitions force an increase in either count, and if so, how does the gap grow with  $n$ ?
4. Can we say something general about the number of states and transitions when we eliminate epsilon transitions from NFAs?